

Lecture 11, Feb 18 2026

7.2 trigonometric integrals.

Example. Evaluate $\int \cos^3 x \, dx$.

Sln.

$$\cos^3 x = \cos x \cdot \cos^2 x = \cos x (1 - \sin^2 x) = \cos x - \cos x \sin^2 x.$$

then

$$\begin{aligned} \int \cos^3 x \, dx &= \int (\cos x - \cos x \sin^2 x) \, dx \\ &= \int \cos x \, dx - \int \cos x \sin^2 x \, dx & u = \sin x \\ &= \sin x - \int u^2 \, du & du = \cos x \, dx \\ &= \sin x - \frac{u^3}{3} + C \\ &= \sin x - \frac{\sin^3 x}{3} + C. \end{aligned}$$

Example. Find $\int \sin^5 x \cos^2 x \, dx$.

Sln. $\sin^5 x \cos^2 x = (\sin^2 x)^2 \cdot \sin x \cos^2 x$.

$$\sin^2 x = 1 - \cos^2 x.$$

$$= (1 - \cos^2 x)^2 \cos^2 x \sin x.$$

$$u = \cos x$$

$$du = -\sin x \, dx.$$

$$\begin{aligned} \int \sin^5 x \cos^2 x \, dx &= \int (1 - \cos^2 x)^2 \cos^2 x \sin x \, dx \\ &= - \int (1 - u^2)^2 u^2 \, du \\ &= - \int (1 - 2u^2 + u^4) u^2 \, du \\ &= - \int (u^2 - 2u^4 + u^6) \, du \\ &= - \frac{u^3}{3} - \frac{2}{5} u^5 + \frac{1}{7} u^7 + C \\ &= - \frac{1}{3} \cos^3 x - \frac{2}{5} \cos^5 x + \frac{1}{7} \cos^7 x + C. \end{aligned}$$

$$\sin^2 x = \frac{1}{2} (1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2} (1 + \cos 2x).$$

Example. Evaluate $\int_0^{\pi} \sin^2 x \, dx$.

Sln.

$$\begin{aligned} \int_0^{\pi} \sin^2 x \, dx &= \frac{1}{2} \int_0^{\pi} (1 - \cos 2x) \, dx = \frac{1}{2} \pi - \frac{1}{4} \sin 2x \Big|_0^{\pi} = \frac{\pi}{2} \\ &= \frac{1}{2} \pi - \frac{1}{4} (\sin 2\pi - \sin 0) \end{aligned}$$

$$\int \cos(\alpha x) dx = \frac{1}{\alpha} \sin \alpha x + C \quad \int \sin \beta x dx = -\frac{1}{\beta} \cos \beta x + C.$$

Example. Find $\int \sin^4 x dx$

Sln.

$$\begin{aligned} \int \sin^4 x dx &= \int (\sin^2 x)^2 dx \\ &= \int \left(\frac{1}{2} (1 - \cos 2x) \right)^2 dx && (x \cdot y)^n = x^n y^n \\ &= \frac{1}{4} \int (1 - 2 \cos 2x + \cos^2 2x) dx \\ &= \frac{1}{4} \int \left(1 - 2 \cos 2x + \frac{1}{2} (1 + \sin 2x) \right) dx \\ &= \frac{1}{4} \int \left(\frac{3}{2} - 2 \cos 2x + \frac{1}{2} \sin 2x \right) dx \\ &= \frac{3}{8} x - \frac{1}{2} \int \cos 2x dx + \frac{1}{8} \int \sin 2x dx \\ &= \frac{3}{8} x - \frac{1}{2} \cdot \frac{1}{2} \sin 2x - \frac{1}{8} \cdot \frac{1}{2} \cos 2x + C \\ &= \frac{3}{8} x - \frac{1}{4} \sin 2x - \frac{1}{16} \cos 2x + C. \end{aligned}$$

Thm. (Strategy for evaluating $\int \sin^m x \cos^n x dx$.)

a). If the power of cosine is odd ($n = 2k+1$), save one cosine factor and use $\cos^2 x = 1 - \sin^2 x$ to express the remaining factors in terms of Sine:

$$\int \sin^m x \cos^{2k+1}(x) dx = \int \sin^m x (\cos^2 x)^k \cdot \cos x dx.$$

$$= \int \sin^m x (1 - \sin^2 x)^k \cdot \cos x dx.$$

$u = \sin x \Rightarrow$ the integral becomes in an integral of some polynomial.

$$= \int u^m (1 - u^2)^k du.$$

b). If the power of sine is odd ($m=2k+1$), save one factor and use $\sin^2 x = 1 - \cos^2 x$ to express the remaining factors in terms of cosine.

$$\begin{aligned}\int \sin^{2k+1} x \cos^n x dx &= \int (\sin^2 x)^k \cos^n x \cdot \sin x dx \\ &= \int (1 - \cos^2 x)^k \cos^n x \cdot \sin x dx\end{aligned}$$

$$u = \cos x \quad du = -\sin x \quad = - \int (1 - u^2)^k u^n du.$$

c). If the powers of both sine and cosine are even, use the half-angle identity

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x).$$

Sometimes

$$\sin x \cos x = \frac{1}{2} \sin 2x.$$

Integrals of Powers of Secant and Tangent.

Example. Evaluate $\int \tan^6 x \sec^4 x dx$.

Sln. $\sec^2 x = 1 + \tan^2 x$.

$$\int \tan^6 x \sec^4 x dx = \int \tan^6 x \sec^2 x \cdot \sec^2 x dx.$$

$$= \int \tan^6 x (1 + \tan^2 x) \cdot \sec^2 x dx.$$

$$u = \tan x$$

$$du = \sec^2 x dx$$

$$= \int u^6 (1 + u^2) du.$$

$$= \int u^6 du + \int u^8 du$$

$$= \frac{1}{7} u^7 + \frac{1}{9} u^9 + C$$

$$= \frac{1}{7} \tan^7 x + \frac{1}{9} \tan^9 x + C.$$

Example: Find $\int \tan^5 \theta \sec^7 \theta d\theta$.

soln. $\int \tan^5 \theta \sec^7 \theta d\theta = \int \tan^4 \theta \sec^6 \theta \sec \theta \tan \theta d\theta$

$$= \int (\tan^2 \theta)^2 \sec^6 \theta \cdot \sec \theta \tan \theta d\theta$$

$\tan^2 \theta = \sec^2 \theta - 1 \rightarrow = \int (\sec^2 \theta - 1)^2 \sec^6 \theta \cdot \underbrace{\sec \theta \tan \theta d\theta}_{\substack{u = \sec \theta \\ du = \sec \theta \tan \theta d\theta}}$

$$= \int (u^2 - 1)^2 u^6 du$$

$$= \int (u^4 - 2u^2 - 1) u^6 du$$

$$= \int (u^{10} - 2u^8 - u^6) du$$

$$= \frac{1}{11} u^{11} - \frac{2}{9} u^9 - \frac{1}{7} u^7 + C$$

$$= \frac{1}{11} \sec^{11} \theta - \frac{2}{9} \sec^9 \theta - \frac{1}{7} \sec^7 \theta + C$$

thm (Strategy for evaluating $\int \tan^m x \sec^n x dx$.

a). If the power of secant is even ($n=2k$) save a factor of $\sec^2 x$ and use $\sec^2 x = 1 + \tan^2 x$ to express the remaining factors in terms of $\tan x$.

$$\int \tan^m x \sec^{2k} x dx = \int \tan^m x \cdot (\sec^2 x)^k dx$$

$$= \int \tan^m x (\sec^2 x)^{k-1} \cdot \sec^2 x dx$$

$$u = \tan x$$

$$du = \sec^2 x dx$$

$$= \int u^m (1+u^2)^{k-1} du$$

$$\sec^2 x = 1 + u^2$$

b). If the power of tangent is odd ($m=2k+1$) save a factor of $\sec x \tan x$ and use $\tan^2 x = \sec^2 x - 1$ to express the remaining factors in terms of $\sec x$.

$$\begin{aligned}\int \tan^{2k+1} x \sec^n x dx &= \int (\tan^2 x)^k \sec^{n-1} x \cdot \sec x \cdot \tan x dx \\ &= \int (\sec^2 x - 1)^k \sec^{n-1} x \sec x \tan x dx\end{aligned}$$

$$u = \sec x \quad = \int (u^2 - 1)^k u^{n-1} du.$$

$$du = \tan x \sec x dx$$

Thm: $\int \sec x dx = \ln |\sec x + \tan x| + C.$

Pf.

$$\begin{aligned}\int \sec x dx &= \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx.\end{aligned}$$

$$\text{let } u = \sec x + \tan x \quad du = (\sec x \tan x + \sec^2 x) dx$$

$$= \int \frac{1}{u} du.$$

$$= \ln |u| + C.$$

$$= \ln |\sec x + \tan x| + C.$$

for you. Prove that $\int \csc x dx = \ln |\csc x - \cot x| + C.$

Example: $\int \tan^3 x dx$

Sln: $\int \tan^3 x dx = \int \tan x \cdot \tan^2 x dx = \int \tan x (\sec^2 x - 1) dx$

$$= \int \tan x \sec^2 x dx - \int \tan x dx.$$

$$= \int \sec x \cdot \sec x \cdot \tan x dx - \int \tan x dx$$

$$= \frac{1}{2} \sec^2 x - \int \tan x dx.$$

$$= \frac{1}{2} \sec^2 x - \ln |\sec x| + C.$$

Example. Find $\int \sec^3 x dx$

Sln. $u = \sec x \quad dv = \sec^2 x dx$

$$du = \sec x \tan x \quad v = \tan x$$

By IP.

$$\int \sec^3 x dx = \sec x \tan x - \int \sec x \cdot \tan^2 x dx$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) dx.$$

$$= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx$$

Implies

$$2 \int \sec^3 x dx = \sec x \tan x + \int \sec x dx$$

$$\int \sec^3 x = \frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + C.$$

Example. Evaluate $\int \sin 4x \cos 5x dx$.

Sln. $\sin A \cos B = \frac{1}{2} [\sin(A-B) + \sin(A+B)]$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$A = 4x \quad B = 5x$$

$$\int \sin 4x \cos 5x dx = \frac{1}{2} \int (\sin(-x) + \sin(9x)) dx$$

$$= -\frac{1}{2} \int \sin x dx + \frac{1}{2} \int \sin(9x) dx$$

$$= \frac{1}{2} \cos x - \frac{1}{18} \cos 9x + C.$$

$$\int \cot^m x \csc^n x dx \quad \text{using} \quad 1 + \cot^2 x = \csc^2 x.$$